

Proof Tree

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Algebra

Group Theory

Group

A *group* is a set G with an operation $\cdot$ such that the following properties (often called the *Axioms of Group Theory*) hold:

- Closure: $\forall a, b \in G, a \cdot b \in G$
- Associativity: $\forall a, b, c \in G, a \cdot (b \cdot c) = (a \cdot b) \cdot c$
- Identity: $\exists e \in G$ such that $\forall a \in G, a \cdot e = e \cdot a = a$
- Inverse: $\forall a \in G, \exists a^{-1}$ such that $aa^{-1} = a^{-1}a = e$

Explanation

Proof(s)

History

Applications

Dependencies

Dependents

- Group Identity Is Unique
- Order in a Group
- Abelian Group
- Group Cancellation

Sources

Abelian Group

A G is *abelian* iff $\forall a, b \in G, a \cdot b = b \cdot a$.

Explanation

Proof(s)

History

Applications

Dependencies

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Dependents

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Set Theory

Topology

Analysis

Real Analysis

Geometry